Sparse Alpha-Aware Error-State IEKF for Dynamic 3D Gaussian Splatting

A compact theoretical formulation from nonlinear rendering to state estimation

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Purpose. This note isolates the theory of a state-estimation formulation for dynamic 3D Gaussian Splatting (3DGS). Ground-truth multiview RGB-D images at frame k are treated as observations of a dynamical system whose latent state is the pose and motion of the Gaussians. The renderer is the nonlinear measurement model. Its Jacobian is derived analytically from projection, covariance projection, Gaussian footprint evaluation, and front-to-back alpha compositing. The resulting measurement Jacobian is sparse because each pixel depends only on the Gaussians that contribute to its ray/tile. The state correction is obtained from an error-state iterated extended Kalman filter (IEKF), preferably in information form, rather than from generic loss backpropagation and Adam.

Central design principle.

$$\boxed{\text{analytic measurement model} \;+\; \text{analytic local Jacobian} \;+\; \text{probabilistic estimator}}$$

not

$$\boxed{\text{render loss} \;\longrightarrow\; \text{automatic differentiation} \;\longrightarrow\; \text{generic optimizer.}}$$

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1 Problem setting

Let the canonical 3DGS contain N Gaussians. Gaussian i has canonical appearance and intrinsic shape

$$\mathcal{G}_i^0 = \big(c_i, o_i, \Lambda_i\big), \qquad \Lambda_i = \mathrm{diag}(\sigma_{i1}^2, \sigma_{i2}^2, \sigma_{i3}^2),$$

and time-varying geometry

$$\mu_{i,k} \in \mathbb{R}^3, \qquad R_{i,k} \in \mathrm{SO}(3).$$

Throughout the initial estimation stage, appearance is frozen:

$$\boxed{\dot{c}_i = 0, \qquad \dot{o}_i = 0, \qquad \dot{\Lambda}_i = 0.} \tag{1}$$

Thus every change in rendered RGB-D is attributed to geometry, visibility, occlusion, or model mismatch, not to trainable appearance adaptation.

At discrete frame k, the known camera set is $\mathcal{C}_k$. Camera c has calibrated intrinsics K_c and known world-to-camera transform $(R_{cw,c}, t_{cw,c})$. The observation is the stacked multiview RGB-D image

$$y_k = \{I_{k,c}^{\mathrm{GT}}, D_{k,c}^{\mathrm{GT}}\}_{c \in \mathcal{C}_k}. \tag{2}$$

The state evolves as

$$x_{k+1} = f_k(x_k) + w_k, \qquad w_k \sim \mathcal{N}(0, Q_k), \tag{3}$$

and the image is generated by the nonlinear renderer

$$\boxed{y_k = h_k(x_k) + n_k, \qquad n_k \sim \mathcal{N}(0, R_k).} \tag{4}$$

The task is to infer x_k recursively from the sequence of observations.

2 State and error-state representation

2.1 Per-Gaussian dynamic state

Before reducing the motion to rigid bodies or joints, a useful bootstrap state is

$$x_{i,k} = \big(\mu_{i,k}, R_{i,k}, v_{i,k}, \omega_{i,k}\big), \tag{5}$$

where v_i and ω_i are world-frame linear and angular velocities. The full state is

$$x_k = (x_{1,k}, \ldots, x_{N,k}).$$

A minimal local perturbation uses a 12-dimensional error state

$$\boxed{\delta x_{i,k} = \begin{bmatrix} \delta\mu_{i,k} \\ \delta\phi_{i,k} \\ \delta v_{i,k} \\ \delta\omega_{i,k} \end{bmatrix} \in \mathbb{R}^{12}.} \tag{6}$$

The retraction is

$$\mu_i^+ = \mu_i^- + \delta\mu_i, \tag{7}$$
$$R_i^+ = \mathrm{Exp}([\delta\phi_i]_\times)\, R_i^-, \tag{8}$$
$$v_i^+ = v_i^- + \delta v_i, \tag{9}$$
$$\omega_i^+ = \omega_i^- + \delta\omega_i. \tag{10}$$

The use of a multiplicative rotation error is essential: R_i is constrained to SO(3) and is not updated as an unconstrained Euclidean vector.

2.2 Discrete constant-twist process model

A neutral process model, appropriate before a joint model is known, is

$$\mu^-_{i,k+1} = \mu^+_{i,k} + v^+_{i,k}\Delta t_k, \tag{11}$$

$$R^-_{i,k+1} = \mathrm{Exp}\Big([\omega^+_{i,k}]_\times \Delta t_k\Big)\, R^+_{i,k}, \tag{12}$$

$$v^-_{i,k+1} = v^+_{i,k}, \tag{13}$$

$$\omega^-_{i,k+1} = \omega^+_{i,k}. \tag{14}$$

For the translational blocks, the first-order error transition is exactly

$$\begin{bmatrix}\delta\mu_{k+1}\\ \delta v_{k+1}\end{bmatrix} = \begin{bmatrix}I & \Delta t I\\ 0 & I\end{bmatrix}\begin{bmatrix}\delta\mu_k\\ \delta v_k\end{bmatrix} + w. \tag{15}$$

For rotation, define $\varphi = \omega\Delta t$ and $A = \mathrm{Exp}([\varphi]_\times)$. Under a left-multiplicative orientation error, first-order propagation gives

$$\boxed{\delta\phi_{k+1} \simeq A\,\delta\phi_k + J_r(\varphi)\,\Delta t\,\delta\omega_k,} \tag{16}$$

where J_r is the right Jacobian of SO(3). Equation (16) supplies the rotational blocks of the process Jacobian F_k.

The predicted covariance obeys the usual EKF relation

$$P^-_{k+1} = F_k P^+_k F_k^\mathsf{T} + Q_k. \tag{17}$$

In the large-scale formulation, this covariance is represented approximately through sparse/block structure or its information operator rather than as a dense matrix.

3 Nonlinear Gaussian-to-pixel measurement model

This section defines $h_k(x_k)$ explicitly.

3.1 World-to-camera geometry

For one camera, omit the camera index. The Gaussian center in camera coordinates is

$$m_i = R_{cw}\mu_i + t_{cw} = \begin{bmatrix}X_i & Y_i & Z_i\end{bmatrix}^\mathsf{T}. \tag{18}$$

Perspective projection gives

$$\boxed{u_i = \pi(m_i) = \begin{bmatrix}f_x X_i/Z_i + c_x\\ f_y Y_i/Z_i + c_y\end{bmatrix}.} \tag{19}$$

Its Jacobian is

$$\boxed{J_{\pi,i} = \frac{\partial\pi}{\partial m_i} = \begin{bmatrix}\frac{f_x}{Z_i} & 0 & -\frac{f_x X_i}{Z_i^2}\\ 0 & \frac{f_y}{Z_i} & -\frac{f_y Y_i}{Z_i^2}\end{bmatrix}.} \tag{20}$$

3.2 3D and projected covariance

The world covariance is

$$\Sigma_i = R_i \Lambda_i R_i^\mathsf{T}, \tag{21}$$

and in camera coordinates

$$\Sigma^c_i = R_{cw}\Sigma_i R_{cw}^\mathsf{T}. \tag{22}$$

The projected 2D covariance is approximated by the local perspective map

$$\boxed{S_i = J_{\pi,i}\Sigma_i^c J_{\pi,i}^{\mathsf{T}} + \varepsilon_s I_2,} \tag{23}$$

where $\varepsilon_s > 0$ denotes any renderer regularization/minimum footprint term. Define the 2D conic

$$Q_i = S_i^{-1}. \tag{24}$$

3.3 Pixel footprint and alpha

For pixel center $x_p \in \mathbb{R}^2$ define

$$e_{pi} = x_p - u_i. \tag{25}$$

Ignoring implementation-specific alpha clamps inside a smooth active region, the Gaussian opacity at the pixel is

$$\boxed{\alpha_{pi} = o_i \exp\left(-\frac{1}{2} e_{pi}^{\mathsf{T}} Q_i e_{pi}\right).} \tag{26}$$

This is already nonlinear in (μ_i, R_i) through u_i, $J_{\pi,i}$, S_i and Q_i.

3.4 Front-to-back compositing

Let $\mathcal{S}_p = (i_1, \ldots, i_{M_p})$ be the contributors to pixel p, sorted front-to-back. For readability re-index them as $i = 1, \ldots, M_p$. Define

$$T_{p1} = 1, \tag{27}$$

$$T_{p,i+1} = T_{pi}(1 - \alpha_{pi}), \tag{28}$$

$$w_{pi} = T_{pi}\alpha_{pi}. \tag{29}$$

The RGB measurement is

$$\boxed{C_p = \sum_{i=1}^{M_p} w_{pi} c_i + T_{p,M_p+1} c_{\mathrm{bg}}.} \tag{30}$$

Define accumulated alpha and depth numerator

$$A_p = \sum_i w_{pi}, \tag{31}$$

$$Z_p = \sum_i w_{pi} d_i, \tag{32}$$

where $d_i = e_3^{\mathsf{T}} m_i$ is the center depth. The normalized depth prediction is

$$\boxed{D_p = \frac{Z_p}{A_p + \varepsilon_D}.} \tag{33}$$

For an RGB-D pixel measurement

$$y_p = \begin{bmatrix} C_p^{\mathsf{T}} & D_p \end{bmatrix}^{\mathsf{T}} \in \mathbb{R}^4, \tag{34}$$

Equations (19)–(33) define the nonlinear measurement map

$$\boxed{y_p = h_p(\{\mu_i, R_i\}_{i \in \mathcal{S}_p}).} \tag{35}$$

4 Analytic local linearization

The IEKF requires the local relation

$$\boxed{\delta y_p \simeq \sum_{i \in \mathcal{S}_p} H_{p,i}\, \delta\xi_i, \qquad \delta\xi_i = \begin{bmatrix} \delta\mu_i \\ \delta\phi_i \end{bmatrix} \in \mathbb{R}^6,} \tag{36}$$

with

$$H_{p,i} = \frac{\partial y_p}{\partial \delta\xi_i} \in \mathbb{R}^{4\times 6}.$$

Velocity columns of the instantaneous image Jacobian are zero; velocity is corrected indirectly through process-state cross-correlation across time.

4.1 Center perturbation

A center perturbation produces

$$\delta m_i = R_{cw}\delta\mu_i, \qquad \boxed{\delta u_i = J_{\pi,i} R_{cw}\delta\mu_i.} \tag{37}$$

Because J_π depends on m_i, its differential is nonzero. For $\delta m = (\delta X, \delta Y, \delta Z)$,

$$\delta J_\pi = \begin{bmatrix} -\frac{f_x}{Z^2}\delta Z & 0 & -\frac{f_x}{Z^2}\delta X + \frac{2f_x X}{Z^3}\delta Z \\ 0 & -\frac{f_y}{Z^2}\delta Z & -\frac{f_y}{Z^2}\delta Y + \frac{2f_y Y}{Z^3}\delta Z \end{bmatrix}. \tag{38}$$

4.2 Rotation perturbation and covariance differential

With a left perturbation $R_i^+ = \mathrm{Exp}([\delta\phi_i]_\times) R_i$, first order gives

$$\boxed{\delta\Sigma_i = [\delta\phi_i]_\times \Sigma_i - \Sigma_i [\delta\phi_i]_\times.} \tag{39}$$

Thus

$$\delta\Sigma_i^c = R_{cw}\delta\Sigma_i R_{cw}^{\mathsf{T}}. \tag{40}$$

The projected covariance differential is

$$\boxed{\delta S_i = (\delta J_\pi)\Sigma_i^c J_\pi^{\mathsf{T}} + J_\pi \delta\Sigma_i^c J_\pi^{\mathsf{T}} + J_\pi \Sigma_i^c (\delta J_\pi)^{\mathsf{T}}.} \tag{41}$$

Since $Q_i = S_i^{-1}$,

$$\boxed{\delta Q_i = -Q_i(\delta S_i)Q_i.} \tag{42}$$

Equation (41) contains both effects: center motion changes the perspective Jacobian, while Gaussian rotation changes the 3D covariance orientation.

4.3 Alpha differential

From (26), with $\delta e_{pi} = -\delta u_i$,

$$\boxed{\delta\alpha_{pi} = \alpha_{pi}\left[e_{pi}^{\mathsf{T}} Q_i \delta u_i - \frac{1}{2} e_{pi}^{\mathsf{T}} (\delta Q_i) e_{pi} \right].} \tag{43}$$

This is the key local relation between a six-dimensional Gaussian pose perturbation and its change in per-pixel alpha.

4.4 Exact tangent alpha-compositing recursion

For a fixed contributor set and fixed depth ordering, differentiate (28):

$$\boxed{\delta T_{p1} = 0, \qquad \delta T_{p,i+1} = (1-\alpha_{pi})\delta T_{pi} - T_{pi}\delta\alpha_{pi}.} \tag{44}$$

Then

$$\boxed{\delta w_{pi} = \alpha_{pi}\delta T_{pi} + T_{pi}\delta\alpha_{pi}.} \tag{45}$$

Because appearance is frozen, $\delta c_i = 0$. Hence

$$\boxed{\delta C_p = \sum_i c_i \delta w_{pi} + c_{\mathrm{bg}}\delta T_{p,M_p+1}.} \tag{46}$$

For depth,

$$\delta A_p = \sum_i \delta w_{pi}, \tag{47}$$

$$\delta d_i = e_3^{\mathsf{T}} R_{cw}\delta\mu_i, \tag{48}$$

$$\delta Z_p = \sum_i \left(d_i\delta w_{pi} + w_{pi}\delta d_i\right), \tag{49}$$

and therefore

$$\boxed{\delta D_p = \frac{\delta Z_p - D_p\delta A_p}{A_p + \varepsilon_D}.} \tag{50}$$

Equations (37)–(50) provide a complete analytic directional derivative of the RGB-D pixel with respect to the Gaussian pose perturbations, without differentiating a scalar loss.

4.5 Equivalent behind-composite interpretation

For RGB, define B_{pi} as the color obtained by compositing all contributors strictly behind Gaussian i, normalized to unit incoming transmittance at i. Then

$$\boxed{\frac{\partial C_p}{\partial \alpha_{pi}} = T_{pi}(c_i - B_{pi}).} \tag{51}$$

This identity clarifies observability: an alpha change is informative only when the Gaussian is visible ($T_{pi} > 0$) and differs radiometrically from what it occludes ($c_i \neq B_{pi}$). Equation (51) is equivalent to the tangent recursion for a single-alpha perturbation; the recursion remains preferable when many alphas vary simultaneously.

5 Sparse global measurement Jacobian

Stack all pixel residuals into

$$r_k = y_k^{\mathrm{GT}} - h_k(\hat{x}_k^-). \tag{52}$$

The first-order model is

$$\boxed{r_k = H_k\delta x_k + n_k.} \tag{53}$$

The key structural fact is

$$\boxed{H_{p,i} = 0 \qquad \text{whenever } i \notin \mathcal{S}_p.} \tag{54}$$

Thus H is the incidence operator of a bipartite pixel–Gaussian visibility graph. Every row touches only the state blocks of the Gaussian contributors of that pixel.

Consequently, the block (i, j) of the measurement information matrix

$$\mathcal{I}_k^{\text{meas}} = H_k^\top R_k^{-1} H_k \tag{55}$$

is nonzero only if Gaussian i and Gaussian j are co-visible in at least one measurement pixel:

$$\boxed{[\mathcal{I}_k^{\text{meas}}]_{ij} \neq 0 \implies \exists p : \; i, j \in \mathcal{S}_p.} \tag{56}$$

Occlusion therefore induces structured off-diagonal information rather than independent per-Gaussian updates.

6 From the linearized renderer to the EKF estimator

6.1 Linearized Bayesian update

At frame k, let the predicted error state satisfy

$$\delta x_k \sim \mathcal{N}(0, P_k^-), \qquad \Lambda_k^- = (P_k^-)^{-1}. \tag{57}$$

The linearized measurement is (53). The posterior MAP correction minimizes

$$\boxed{\mathcal{J}_k(\delta x) = \frac{1}{2}\delta x^\top \Lambda_k^- \delta x + \frac{1}{2}(r_k - H_k \delta x)^\top R_k^{-1}(r_k - H_k \delta x).} \tag{58}$$

Setting the gradient to zero gives the information-form normal equations

$$\boxed{\left(\Lambda_k^- + H_k^\top R_k^{-1} H_k\right) \delta x_k = H_k^\top R_k^{-1} r_k.} \tag{59}$$

Equation (59) is algebraically equivalent to the standard EKF gain formula

$$K_k = P_k^- H_k^\top (H_k P_k^- H_k^\top + R_k)^{-1}, \qquad \delta x_k = K_k r_k, \tag{60}$$

but is preferable conceptually and computationally when the measurement dimension is enormous and H is sparse.

The local posterior information is

$$\boxed{\Lambda_k^+ \simeq \Lambda_k^- + H_k^\top R_k^{-1} H_k,} \tag{61}$$

and the Gaussian approximation to the posterior covariance is $(\Lambda_k^+)^{-1}$ on the represented state subspace.

6.2 Why this differs from gradient descent

For a least-squares image loss, generic gradient descent uses essentially

$$-\nabla_x L \propto H^\top R^{-1} r. \tag{62}$$

The estimator instead solves (59), which additionally uses

$$\boxed{\Lambda^- + H^\top R^{-1} H.} \tag{63}$$

Thus the update depends on prior uncertainty, measurement uncertainty, local observability, parameter coupling, and pixel–Gaussian co-visibility. There is no arbitrary learning rate in the ideal linear-Gaussian update.

7 Iterated EKF for the nonlinear renderer

A single linearization can be inaccurate because $h(x)$ is nonlinear. Let x_k^- be the fixed process prior for the current frame. Introduce tangent coordinates δ through

$$x(\delta) = x_k^- \boxplus \delta. \tag{64}$$

The exact framewise MAP objective is

$$\boxed{\mathcal{F}_k(\delta) = \frac{1}{2}\|\delta\|^2_{\Lambda_k^-} + \frac{1}{2}\|y_k - h_k(x_k^- \boxplus \delta)\|^2_{R_k^{-1}}.} \tag{65}$$

At IEKF/Gauss–Newton iteration ℓ, let δ_ℓ be the current tangent estimate, $x_\ell = x^- \boxplus \delta_\ell$, residual

$$r_\ell = y - h(x_\ell), \tag{66}$$

and measurement Jacobian H_ℓ. Linearizing the residual in an increment Δ gives

$$\boxed{\left(\Lambda^- + H_\ell^\mathsf{T} R^{-1} H_\ell + \lambda_\ell D_\ell\right)\Delta_\ell = H_\ell^\mathsf{T} R^{-1} r_\ell - \Lambda^- \delta_\ell.} \tag{67}$$

Here $\lambda_\ell D_\ell \succeq 0$ is a Levenberg–Marquardt regularization. The update is accepted with a line-search factor $\alpha_\ell \in (0, 1]$:

$$\delta_{\ell+1} = \delta_\ell + \alpha_\ell \Delta_\ell, \tag{68}$$

provided the true nonlinear objective (65) decreases. The same observation is never treated as multiple independent measurements; relinearization refines one MAP update with the same fixed prior.

8 Piecewise smoothness and the active visibility set

The formulas above assume that, during one local linearization, the following discrete renderer structure is fixed:

- the set $\mathcal{S}_p$ of contributors to each pixel/tile;
- their front-to-back depth order;
- the active branch of any alpha clamp or support threshold.

Under this fixed active set, $h(x)$ is smooth in a neighborhood of the linearization point. A sufficiently large state update can change sorting, support, or clamping and invalidate the old Jacobian. Therefore the IEKF must be understood as a *piecewise-smooth* estimator: after an accepted nonlinear state update, the renderer is evaluated again and the visibility structure is rebuilt before the next linearization.

9 Observability and information

9.1 Instantaneous pixel information

For Gaussian i, the instantaneous pose information contributed by the current frame is

$$\boxed{\mathcal{I}_{i,k} = \sum_{p:\, i \in \mathcal{S}_p} H_{p,i}^\mathsf{T} R_p^{-1} H_{p,i}.} \tag{69}$$

Its rank reveals which pose directions are locally observable. Examples:

- a fully occluded Gaussian has $H_{p,i} = 0$ and therefore no visual information;
- rotation of a perfectly isotropic Gaussian ($\Lambda_i \propto I$) does not change Σ_i, so its orientation is unobservable from footprint geometry;
- depth strongly constrains translation along the camera optical direction when the Gaussian contributes appreciably to the pixel;
- multiple calibrated views increase rank by observing different projections of the same motion.

9.2 Dynamic observability across frames

Even though an image does not directly depend on v_i or ω_i, the process model couples velocity to future pose. For a linearized time-varying system

$$\delta x_{k+1} = F_k \delta x_k + w_k, \qquad r_k = H_k \delta x_k + n_k, \tag{70}$$

define the transition

$$\Phi_{j,k} = F_{j-1} \cdots F_k, \qquad j > k. \tag{71}$$

The finite-horizon observability information is

$$\boxed{\mathcal{O}_{k,L} = \sum_{j=k}^{k+L} \Phi_{j,k}^{\mathsf{T}} H_j^{\mathsf{T}} R_j^{-1} H_j \Phi_{j,k}.} \tag{72}$$

Velocity and angular velocity become estimable when (72) has sufficient rank on their subspace. This is the formal reason why image-based pose evidence can infer velocity without interpreting a photometric gradient as a physical velocity.

10 Local convergence of the damped IEKF

Assumption 1. *Within a neighborhood of a solution and a fixed renderer active set: (i) h is continuously differentiable and its Jacobian is locally Lipschitz; (ii) $R \succ 0$ and $\Lambda^- \succ 0$ on the represented error-state subspace; (iii) the relevant level set of $\mathcal{F}$ is bounded; and (iv) each accepted step satisfies a sufficient-decrease line-search condition.*

Define the Gauss–Newton/LM matrix

$$B_\ell = \Lambda^- + H_\ell^{\mathsf{T}} R^{-1} H_\ell + \lambda_\ell D_\ell. \tag{73}$$

For $\lambda_\ell D_\ell \succeq 0$,

$$B_\ell \succ 0. \tag{74}$$

Let $g_\ell = \nabla \mathcal{F}(\delta_\ell)$ and $\Delta_\ell = -B_\ell^{-1} g_\ell$ (equivalent to (67) in the chosen tangent coordinates). Then, for $g_\ell \neq 0$,

$$\boxed{g_\ell^{\mathsf{T}} \Delta_\ell = -g_\ell^{\mathsf{T}} B_\ell^{-1} g_\ell < 0.} \tag{75}$$

Thus the LM/IEKF direction is a strict descent direction.

Proposition 1 (Monotone framewise convergence)**.** *Under Assumption 1, an Armijo/backtracking line search applied to the damped IEKF direction produces*

$$\mathcal{F}(\delta_{\ell+1}) \leq \mathcal{F}(\delta_\ell), \tag{76}$$

and since $\mathcal{F} \geq 0$, the objective values converge. Under the standard boundedness conditions on B_ℓ and local Lipschitz continuity of $\nabla \mathcal{F}$, every accumulation point is stationary and

$$\|\nabla \mathcal{F}(\delta_\ell)\| \to 0. \tag{77}$$

This is a local optimization-convergence statement, not a claim of global physical identifiability. Convergence to the true state additionally requires sufficient observability, a correct measurement/process model, bounded noise, and an initialization in the basin of the desired solution.

11 Sparse information form and matrix-free interpretation

Equation (59) contains the large operator

$$A = \Lambda^- + H^\mathsf{T} R^{-1} H. \tag{78}$$

The theory does not require explicit construction of H. For any vector v,

$$\boxed{Av = \Lambda^- v + H^\mathsf{T} R^{-1}(Hv).} \tag{79}$$

Hence the estimator only requires the two linear operators

$$\text{JVP: } v \mapsto Hv, \qquad \text{VJP: } z \mapsto H^\mathsf{T} z. \tag{80}$$

The right-hand side is simply

$$b = H^\mathsf{T} R^{-1} r. \tag{81}$$

The per-Gaussian block information

$$M_i = \Lambda^-_{ii} + \sum_{p: i \in \mathcal{S}_p} H_{p,i}^\mathsf{T} R_p^{-1} H_{p,i} \tag{82}$$

is a natural local approximation to A and also a direct measure of Gaussian observability.

The sparsity is not an approximation to the renderer: it follows exactly from the finite contributor relation (54). Any additional approximation enters only through how the prior information and long-range cross-correlations are represented.

12 Reduction after rigid/joint structure emerges

The high-dimensional Gaussian state is useful as a bootstrap representation. Once a rigid motion model is identified, the same measurement Jacobian can be projected analytically onto the physically admissible subspace.

For a prismatic joint with unit axis e,

$$\frac{\partial \mu_i}{\partial q} = e, \qquad \frac{\partial \phi_i}{\partial q} = 0. \tag{83}$$

For a revolute joint with unit axis e through point a,

$$\boxed{\frac{\partial \mu_i}{\partial q} = e \times (\mu_i - a), \qquad \frac{\partial \phi_i}{\partial q} = e.} \tag{84}$$

Thus the pixel-to-joint Jacobian is

$$\boxed{H_{p,q} = \sum_{i \in \mathcal{S}_p} H_{p,i} \begin{bmatrix} \partial \mu_i / \partial q \\ \partial \phi_i / \partial q \end{bmatrix}.} \tag{85}$$

The same nonlinear renderer is retained, but the estimator state collapses from thousands of Gaussian pose variables to the physically meaningful coordinate q (and, if needed, $\dot{q}$ and a small set of joint parameters). Equation (85) is the formal bridge between Gaussian-level bootstrap estimation and joint-level state estimation.

13 Summary of the theoretical construction

The complete argument can be compressed into the following chain:

State:	$x_k = \{\mu_i, R_i, v_i, \omega_i\}_{i=1}^N$,
Process:	$x_{k+1} = f_k(x_k) + w_k$,
Nonlinear measurement:	$y_k = h_k(x_k) + n_k$,
Renderer geometry:	$(\mu_i, R_i) \to (u_i, Q_i) \to \alpha_{pi} \to T_{pi} \to (C_p, D_p)$,
Local analytic linearization:	$r_k \simeq H_k \delta x_k + n_k$,
Sparsity:	$H_{p,i} = 0 \quad \text{for } i \notin \mathcal{S}_p$,
Linearized MAP/EKF update:	$(\Lambda^- + H^\mathsf{T} R^{-1} H)\delta x = H^\mathsf{T} R^{-1} r$,
Nonlinear correction:	relinearized, damped IEKF with true-objective line search,
Physical reduction:	$H_q = H_G \, \partial G / \partial q$.

The essential distinction from generic differentiable 3DGS training is that the analytic derivative is a derivative of the *measurement function* h, not of an arbitrary scalar training loss. The estimator then combines this Jacobian with uncertainty and dynamics. This preserves the geometric meaning of the renderer, exposes observability explicitly, and makes the transition from per-Gaussian state estimation to rigid/joint estimation mathematically direct.